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Carbon-tax implementation in Indonesia: a social accounting matrix analysis

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ABSTRACT

As the adverse effects of climate change intensify, numerous countries are making several efforts to reduce their carbon-dioxide (CO₂) emissions. One of the widely adopted strategies is the implementation of a carbon tax. This study analyzed the potential impact of a carbon tax on fuel usage on sectoral price changes and distributional impacts on households' living expenses. Three types of simulations were employed: levying the tax only on the electricity sector, the top-10 emitter sectors, and all sectors. Using the Indonesian Social Accounting Matrix (SAM) 2015, as well as the price-multiplier matrix in the simulations, the results show that, of all production sectors, the carbon tax had the most substantial impact on the electricity sector, followed by energy-intensive sectors and rail transportation. We found that the impact of the carbon tax is more detrimental to urban households than rural households. In addition, regressiveness is stronger in urban households than in rural households. Different tax scenarios have varying impacts. Taxing all sectors leads to significant price increases across all sectors and higher living and labor costs for all groups. Government decisions on emissions taxes should carefully consider their economic effects on essential industries and vulnerable populations.

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Carbon tax; price increase; distributional impact; regressive; CO₂ emissions

Introduction

The increasing level of greenhouse-gas (GHG) concentrations is considered to be the main factor causing climate change (Liu, Gao, and Xiao 2019). Compared to other GHG components, carbon dioxide (CO₂) accounts for the largest proportion, making it the leading cause of climate change (IPCC 2014). Along with the worsening conditions due to climate change, all countries are experiencing negative multidimensional impacts on a more frequent basis. Previous literature identifies that climate change causes hydro-meteorological disasters (Veenema et al. 2017), physical and socioeconomic damage (Arouri, Nguyen, and Ben Youssef 2015), and exacerbates various health issues (Su 2011; Alderman, Turner, and Tong 2012; Lane et al. 2013; Gao et al. 2016; Barnes 2018). If no further efforts can halt climate change, massive externality costs will be imposed on future generations. In Indonesia, for example, gross domestic product (GDP) could decline by up to 3.45% by 2050, with the poorest communities disproportionately affected and future generations increasingly vulnerable to extreme

weather, pollution, and natural disasters (UNICEF 2023).

These substantial costs prompted countries to sign the Paris Agreement in 2016 with the aim of mitigating climate change. This accord is an international treaty that legally binds countries to take action to reduce emissions, with the goal of limiting the global average temperature increase to below 2°C compared to pre-industrial levels. Strengthening a nation's resilience involves its ability to adapt to the impacts of climate change, improving climate resilience, and promoting low-carbon development. Additionally, developed countries pledged to create financial frameworks to support low-carbon and climate-resilient development.

Indonesia plays a crucial role in slowing the climate-change process under this agreement. In 2022, the government released its enhanced National Determined Contribution (NDC), which is committed to achieving a 31.98% reduction in CO₂ emissions without international support or a 43.2% reduction with international support by 2030. As one of the world's highest-emitting countries,

Indonesia has also developed a National Energy General Plan (*Rencana Umum Energi Nasional*, or RUEN) to control energy intensity and initiate the energy transition by increasing the share of new renewable energy (NRE) in its energy mix. Several other energy policies have also been implemented, such as the removal of various energy subsidies, the feed-in-tariff implementation in NRE investment, and NRE development. However, as with other coal-intensive countries, these policies are not enough to attain the target as long as decarbonization in the energy sector is not carried out (GEA 2017).

In recent years, Indonesia has been considering the implementation of a significant policy to combat emissions: a carbon tax. A carbon tax is a policy tool designed to reduce GHG emissions by placing a price on carbon emissions. Various countries, including Mexico, Colombia, Chile, Iceland, Denmark, Ireland, the UK, Norway, Poland, Switzerland, India, and Japan, have already implemented this Pigouvian tax (Partnership for Market Readiness 2017). While it is intended to address the pressing issue of climate change, there are valid concerns that it could lead to economic disruptions, such as higher production costs and an increase in the cost of living for low-income households that are already vulnerable.

The unintended consequences of a carbon tax suggest the importance of conducting a comprehensive ex-ante analysis to assess both the direct and indirect socioeconomic impacts of its implementation. Numerous studies have explored the economic repercussions of carbon taxes, primarily focusing on macroeconomic indicators (Meng, Siriwardana, and McNeill 2013; Benavente and Miguel 2016; Nurdianto and Resosudarmo 2016; Fullerton and Muehlegger 2019). However, recent studies have devoted attention to welfare issues to design better measures that could reduce the negative impact of carbon tax. For instance, researchers have examined its impact on commodity prices (Jia and Lin 2020; Lin and Jia 2020) and the distributional effects on household welfare (Feng et al. 2018; Goulder et al. 2019; Jia, Wen, and Liu 2022). The issue of distributional effects has garnered increasing attention in recent literature, as studies have revealed significant variations in impact across sectoral prices, household income levels, and geographic regions. For example, Goulder et al. (2019) has argued that the implementation of a carbon tax in the United States would predominantly affect the affluent class, albeit with certain limitations. In contrast, Jia, Wen, and Liu (2022) found that the introduction of carbon trading in China could have a more substantial impact on rural residents, given their dependence on coal for heating.

In Indonesia, few studies have examined the consequences of implementing a carbon tax. The most recent study by Hartono et al. (2023) uses a dynamic computable general equilibrium (CGE) model to analyze its effects on macroeconomic indicators such as GDP and employment. However, the impacts on distribution and household welfare were last discussed in a 2015 study by Yusuf and Resosudarmo, which used a CGE model to focus on these distributional effects. This study, however, relies on data from 2007, highlighting a significant gap in the current literature regarding the assessment of the distributional impact of carbon taxes in Indonesia.

To address this gap, we examined the impact of carbon tax-induced price increases and the distributional effects among households with different income and geographical areas, providing welfare implications at the level of the detailed household. We utilized Indonesia's social accounting matrix (SAM) framework, known as the *Sistem Neraca Sosial Ekonomi* (SNSE), for the year 2015. The SAM price model is useful for providing a comprehensive overview of the economy, enabling detailed sectoral and distributional analysis of policy impacts while ensuring data consistency. It aids in understanding how economic changes affect different social groups and sectors, making it valuable for targeted policy-making. The data for the SAM are also relatively accessible and have been utilized in recent literature for various environmental studies (Hartono et al. 2020; Llop 2021; Florido and Rosa 2024). However, the model is limited by its static nature, reliance on simplified assumptions, and data intensity. Some variables are assumed to be fixed, which may not capture the full complexity of economic interactions, especially when identifying long-term dynamics. Nevertheless, it is sufficient for analyzing potential short-term impacts. To analyze the effect of price changes, we developed a price-multiplier matrix to examine the overall impacts on sectoral prices resulting from the imposition of a carbon tax. Incorporating prices in the SAM structure allows for absolute price-variation analysis, which differs from the general quantity-based SAM (Roland-Holst and Sancho 1995). The SAM price model has also been used in prior studies identifying the effect of price increases in the food sector (Misdawita and Nugroho 2019) and petroleum prices (Saari, Dietzenbacher, and Los 2016).

This study sets itself apart by employing a simplified CGE and contributes to the literature in several ways. First, it is the first study examining the consequences of implementing a carbon tax on prices through a SAM-based price model in Indonesia. This unique approach allows us to pinpoint the

effects of price increases, which have been largely unexplored in prior research in Indonesia. Notably, price plays a pivotal role in the energy landscape of Indonesia, affecting not only the dynamics of energy supply and demand but also the well-being of households, particularly those in vulnerable situations, which are still prevalent in the country.

This leads us to the second significant contribution: the analysis goes beyond the assessment of the carbon tax's impact solely on commodity prices and production factors. It also extends to evaluating the living costs incurred by households, allowing us to evaluate the social consequences as well. Additionally, we disaggregated household impacts by deciles and regions to analyze the distribution of living costs across different income levels and urban vs. rural areas. The findings demonstrate how the carbon tax affects various income levels and regions and whether its impact is regressive or progressive. This evaluation can provide the government with insights into the potential burden of the carbon tax, enabling the design of more appropriate measures to mitigate these effects. Finally, recognizing the limitations imposed by data availability when conducting a comprehensive assessment such as CGE, our utilization of the SAM framework offers a more straightforward yet valuable tool for policymakers, especially for short-term analysis.

The structure of this article is organized as follows. The next section provides a literature review on carbon-tax adoption and its implications in Indonesia. This is followed by a detailed explanation of the data and methodology, focusing on the use of a SAM-based price model. The results section presents the findings derived from the model, which are further analyzed and interpreted in the discussion section. Finally, the conclusion summarizes the key

findings, addresses limitations, and suggests directions for future research.

Literature review

The following literature review covers two main topics. First, we explore the adoption of carbon taxes in Indonesia, detailing the country's emissions profile, the government's plans for carbon pricing, and comparisons with international practices. Second, this section reviews empirical studies on the implications of carbon taxes using various methodologies, including social accounting matrix (SAM), computable general equilibrium (CGE), and input-output (IO) models.

Carbon-tax adoption in Indonesia

Indonesia emitted 600 million tons of CO₂, making it the world's ninth-largest emitter in 2021 (IEA 2022). While forest and peat-fire emissions accounted for a high share, the energy-sector share rose significantly compared to total national emissions over time (see Figure 1). Thus, if the Indonesian government intends to reduce national CO₂ emissions, it will need to consider the implementation of emission-reduction policies, such as a carbon tax, in the energy sector.

Following the examples of other developing countries, the Indonesian government planned to impose a carbon tax under the fifth amendment to the Tax Law Article 44G in mid-2021. The aim is to impose a tax on entities purchasing goods containing carbon or conducting activities that generate carbon emissions. A carbon tax is one of the carbon-pricing instruments widely used to reduce fossil-energy consumption and start the transition to NRE (World

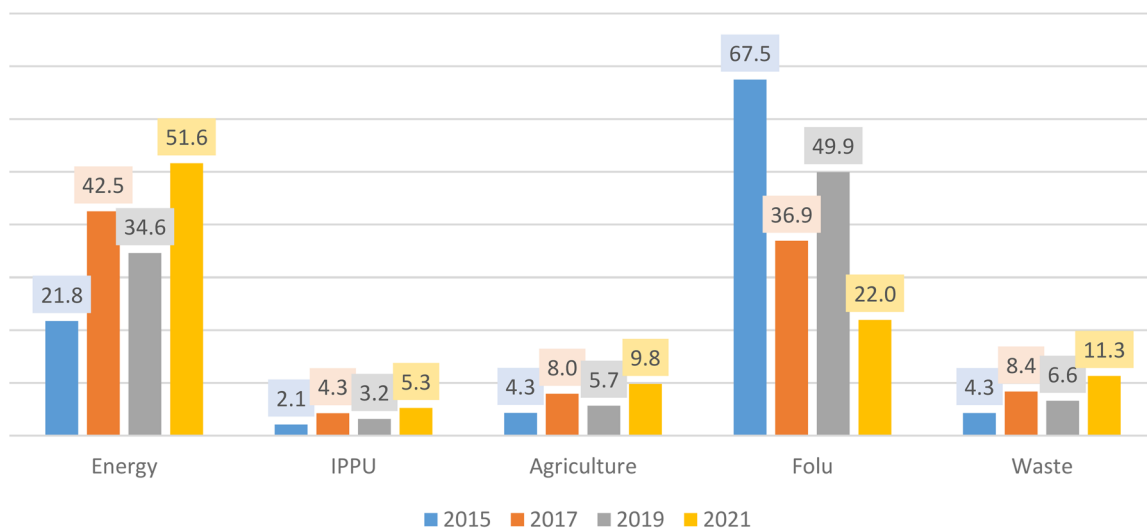


Figure 1. Sectoral emission contribution to Indonesia's national total emissions for 2015–2021 (in %). Source: MEF (2022).

Bank 2019). Carbon pricing is a fiscal policy designed to internalize the external costs of CO₂ emissions. Entities emitting CO₂ can either reduce their emissions or choose to continue emitting by paying for the emissions produced. Another carbon-pricing instrument is the Emission Trading System (ETS), but the ETS is less suitable for implementation in Indonesia due to distorted market conditions and political challenges, particularly in the energy and transportation sectors (Yusuf and Resosudarmo 2015; Kaneko and Kawanishi 2016). A carbon tax is also more appropriate and relatively easy to achieve within Indonesia's existing fiscal administration system. Compared to other high-cost policy instruments, a carbon tax is one of the most efficient emission reduction policy instruments (Tvinnereim and Mehling 2018; Ojha, Pohit, and Ghosh 2020). However, at the same time, the implementation of a carbon tax may shift people's behavior, especially related to the consumption of goods subject to the tax, and potentially lead to unintended consequences.

When implementing a carbon tax, policymakers typically need to consider several key factors. The first is the tax rate, which varies significantly across different initiatives. For instance, carbon pricing ranges from as low as US\$2 per ton in China to over US\$100 per ton in Sweden (Arimura and Matsumoto 2020). Typically, newly established carbon taxes and those in developing countries start at rates no higher than US\$15 per ton of CO₂. In contrast, more established policies in countries like Sweden, Switzerland, and Finland feature significantly higher carbon prices.

Second, the scope of the tax is an important consideration, as carbon taxes can be applied to either upstream or downstream sectors. Generally, carbon taxes are imposed on upstream fossil-fuel sales, such as coal mines, natural gas fields, or oil wells (Khastar, Aslani, and Nejati 2020). The tax results in increased costs for primary fuel consumers, such as electricity providers, oil and gas refineries, and transportation sectors, which then pass these costs onto their final customers—households and businesses. Ultimately, this burden is transferred downstream to end consumers. In cases where the tax is imposed upstream, it can be targeted at specific sectors. For example, Chile and South Africa apply carbon taxes only to the power sector, while China and Norway tax top-emitting sectors such as power and heavy industries. Conversely, countries like Finland and Sweden apply the carbon tax across all sectors.

Third, consideration of complementary policies is essential. Several countries have reported welfare losses, international competitiveness issues, and

employment challenges due to the adoption of a carbon tax. For instance, Finland, despite successfully reducing CO₂ emissions, has experienced adverse effects on social welfare (Khastar, Aslani, and Nejati 2020). Similarly, Japan, which introduced carbon pricing in 2012, is concerned about its impact on international competitiveness and employment while failing to significantly reduce GHG emissions due to a very low carbon price (Arimura and Matsumoto 2020). Additionally, there are reports of potential unequal distributional impacts across households, although the patterns vary across different countries (Dorband et al. 2019; Steckel et al. 2021). Consequently, studies emphasize the importance of identifying additional policies that can mitigate the adverse effects of a carbon tax.

In Indonesia, the government plans to implement a carbon tax of approximately US\$12.00 per ton of emissions. The initial phase will focus on the power sector, which is the largest emitter in the country. If successful, the tax will gradually be expanded to full-scale implementation.

Empirical literature about carbon-tax implications

Carbon-tax impacts have been studied using different methods, yielding varied findings. The methods in previous studies have included input-output (IO) analysis (Gemechu et al. 2014), SAM frameworks (Chapa and Ortega 2017), CGE models (Rausch, Metcalf, and Reilly 2011; Liang and Wei 2012; Yusuf and Resosudarmo 2015), and Quadratic Almost Ideal Demand System (QUAIDS) models (Renner, Lay, and Schleicher 2019; Okonkwo 2021). The QUAIDS model is not used in this study because it provides a detailed representation of the impacts of a carbon tax from a micro-level perspective while the CGE model offers a macro-scale perspective. The econometric approach is infrequently utilized because it concentrates on the effects on specific variables, often overlooking feedback and indirect effects from other sectors. Conversely, partial and general equilibrium models like IO, SAM, and CGE can capture these effects, as they offer an economy-wide analysis. By using equilibrium analyses such as IO and SAM, we find that the carbon tax affects living costs through two channels: changes in production costs and changes in household-purchasing power. Carbon-tax analyses that use a more advanced general equilibrium approach, like the CGE model, provide theoretical and accounting consistencies, while SAM can only provide accounting consistencies. However, complex data preparations are needed when using CGE, including the estimation of elasticity values.

Instead of focusing solely on overall economic indicators like GDP or employment, current discussions have also taken an interest in sub-indicators such as export competitiveness, sectoral prices, household welfare, and distributional impacts across different income classes or regions (Arimura and Matsumoto 2020). One of the various findings regarding the impact of carbon taxes on price increases comes from a study conducted by Chapa and Ortega (2017). These authors found that carbon taxes lead to price increases, which subsequently impact overall welfare levels within communities. Another study by Mathur and Morris (2014), found that a carbon tax leads to higher energy prices and increased prices for other goods and services that rely on energy as an input. This aligns with a study by Gemechu et al. (2014), which concludes that the implementation of a carbon tax in Spain caused a significant increase in production prices for sectors with high emissions and indirectly led to price increases in other sectors.

Regarding the distribution of the impacts, Mathur and Morris (2014) found that a carbon tax is regressive because it affects households in lower deciles more than those in higher deciles. A carbon tax would impose a higher percentage of costs on poor households relative to their expenditures compared to wealthy households, making it a regressive tax (Fremstad and Paul 2019). However, a study by Yusuf and Resosudarmo (2015) using CGE did not find a regressive characteristic of carbon-tax implementation due to the insensitivity of low-income household spending to rising commodity prices related to energy. In addition to its distributional effects on household income, another study by Liang and Wei (2012) reported that the implementation of carbon taxes in China had the potential to increase the prices of goods and services as well as widen the urban-rural gap.

Material and methods

Data

This study uses the 2015 SAM, developed based on the 2015 IO for Indonesia. The construction process involved integrating data from various sources, including the 2015 National Labor Force Survey (SAKERNAS), the 2015 National Socio-Economic Survey (SUSENAS), and additional data from the Central Bureau of Statistics or *Badan Pusat Statistik* (BPS) and the Ministry of Energy and Mineral Resources (MEMR). We used the SUSENAS database to disaggregate households into deciles in both urban and rural areas, following the detailed process outlined in Hartono et al. (2020). To simplify the

Table 1. List of sectors in the 2015 Indonesia SNSE framework.

No.	Sectors
1	Food Crops
2	Other Crops
3	Livestock
4	Forestry
5	Fishery
6	Coal and Metal Mineral
7	Crude Oil and Natural Gas
8	Other Mining
9	Food and Beverage
10	Textile and Leather
11	Timber and Wooden Products
12	Paper
13	Basic non-Ferrous
14	Metal Product
15	Boiler and Prime Mover
16	Electricity Machinery and Equipment
17	Chemical
18	Rubber and Plastic
19	Electricity
20	Gas and Clean Water
21	Power Building Construction
22	Other Construction
23	Trading
24	Restaurant
25	Hotels
26	Rail Transportation
27	Other Land Transportation
28	Air and Water Transportation
29	Transportation Supporting Services
30	Bank and Insurance
31	Real Estate
32	Company Service
33	General Government and Defense
34	Social Community Services
35	Other Services

analysis, the sectors were aggregated into 35 categories, as shown in Table 1. Validation and reconciliation procedures were then conducted to ensure the SAM's balance, as described by Yusuf (2006). A key feature of this SAM is the disaggregation of household-income levels into deciles and the distinction between urban and rural areas, enabling detailed household-level analysis. Additionally, this study processes Indonesia's sectoral energy-consumption data, released by BPS in the 2015 Energy Balances Statistics for Indonesia, along with the intermediate processes of each sector to measure the country's sectoral carbon-emission intensity.

BPS does have more recent SAMs for 2016 and 2022, but this study chose to use the 2015 SAM for two main reasons. First, the 2016 SAM has several limitations. It aggregates sectors into only 17 categories (which is not detailed enough to provide a thorough view of sectors closely related to energy or energy-intensive sectors) and does not differentiate between domestic and imported commodities, which could lead to less accurate SAM multipliers for this study. Additionally, households are not broken down into deciles or separated by urban and rural areas, making it difficult to perform a precise and accurate analysis of household-income distribution across urban and rural households.

Second, the 2022 SAM also has limitations. While it reflects the 2022 economy, its structure follows the 2016 framework, so it does not accurately depict the economic structure of 2022. Moreover, labor as a factor of production is not divided into detailed categories. While households are grouped into quintiles, they are not separated into urban and rural categories, making the income-distribution analysis less effective than the 2015 SAM. Although the 2022 SAM distinguishes between domestic and foreign commodities, the sector classification includes only 17 sectors, which, like the 2016 SAM, is not detailed enough for a thorough analysis of energy-related and energy-intensive sectors, resulting in overly aggregated analysis.

Based on the explanations above, the official 2016 and 2022 BPS SAMs—particularly the 2022 SAM, whose economic structure mirrors that of 2016 and does not accurately reflect 2022—are less suitable for this study compared to the 2015 SAM. Moreover, the one-year difference between the 2015 and 2016 SAMs does not cause significant changes in economic structure. Therefore, we use the 2015 SAM instead of the 2016 or 2022 SAMs due to their aforementioned weaknesses.

Methodology

Building on studies by Roland-Holst and Sancho (1995), Chapa and Ortega (2017), and Renner (2018), this study applies a SAM framework. Specifically, it develops the SAM-based price model to measure the impacts of carbon-tax implementation on price increases across various economic sectors and assesses how these impacts are distributed among different household groups.

SAM is a framework of data arranged into matrices that summarize overall economy transactions and various social economy variables in an integrated manner. It is structured as a 4×4 matrix that represents the flow of money between four main accounts: production activities, factors of production, institutions, and exogenous accounts (such as the capital account and foreign account) (see Figure 3). Production activities, factors, and institutions are categorized as endogenous accounts. In the SAM framework, it is assumed that total expenditure (Y_i) is equal to total income (Y_j), in which Y_j is a transpose matrix of Y_i .

SAM provides a general overview of an economy and the relationships between its variables over a given period. Recognized as a general equilibrium database, SAM encompasses production activities, institutions, and production factors that represent the economy (Pyatt and Round 1979; Defourny and

Thorbecke 1984). It offers a comprehensive view of both social and economic structures within an economy, enabling an understanding of poverty and income distribution (Hartono and Resosudarmo 2008). Furthermore, SAM effectively integrates environmental and economic indicators, facilitating their correlation analysis (Resosudarmo and Thorbecke 1996; Pal, Pohit, and Roy 2012). As a key analytical tool, SAM illustrates the effects of specific economic policies on individual incomes, enabling assessment of their impact on poverty and income distribution. Additionally, SAM's simplicity makes it accessible and adaptable for use across numerous countries (Hartono and Resosudarmo 2008; Endriana et al. 2016).

This study did not use the SAM-based quantity model, which is commonly used in other SAM analyses because the shock in the simulation is the carbon-tax rate (in %) in which the amount depends on the amount of CO₂ emissions in each sector. The carbon-tax rate (in %) for each sector is also different, since it is proportional to the CO₂ emissions produced by each sector and is not proportional to the size of the sectoral output. We used the SAM-based price model, which developed the price multiplier matrix, to capture the impacts of price increases due to the carbon tax on the overall economy sector. The matrix was different from the conventional SAM methodology, which is oriented on quantity and income effects (Roland-Holst and Sancho 1995). The price model was previously developed by Roland-Holst and Sancho (1995), Chapa and Ortega (2017), and Renner (2018) to observe the impacts of an environmental tax on the price of overall economy sectors as well as how the impacts were distributed among various household groups.

Within the SAM framework (see Figure 2), sub-matrix T_{13} illustrates the value added to production factors. Sub-matrix T_{21} shows how income from production factors is allocated to economic institutions, including households, government, and companies, thereby illustrating the distribution of income across these institutions. For example, if some agricultural workers are members of small farmer groups, the money flows from these workers to small farmer households. Sub-matrix T_{22} depicts the transfer of money between institutions, such as government subsidies to households or companies. Sub-matrix T_{32} indicates the demand for goods and services produced by institutions. Finally, sub-matrix T_{33} describes the transactions between production sectors. In the SAM framework (see Figure 2), a key requirement is that total expenditure (Y_i) must equal total income (Y_j),

where Y_j is the transposed result of Y_i and represents each account.

$$\begin{bmatrix} Y_1 \\ Y_2 \\ Y_3 \end{bmatrix} = \begin{bmatrix} 0 & 0 & A_{13} \\ A_{21} & A_{22} & 0 \\ 0 & A_{32} & A_{33} \end{bmatrix} \begin{bmatrix} Y_1 \\ Y_2 \\ Y_3 \end{bmatrix} + \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} \quad (1)$$

The mathematical equation in the SAM framework can be seen in Eq. (1), where X is the input, Y is the output, I is the identity matrix, and A is the average expenditure propensity, which is the sector value divided by the total output. Then, Eq. (1) can be rewritten in Eq. (2).

$$\begin{aligned} Y &= AY + X \leftrightarrow Y - AY = X \leftrightarrow (I - A)Y \\ &= X \leftrightarrow Y = (I - A)^{-1} X \end{aligned} \quad (2)$$

In Eq. (2), matrix $(I - A)^{-1}$ or M_a is the quantity multiplier matrix. Due to the existence of this multiplier matrix, changes in the exogenous account affect the endogenous account. The role of the multiplier matrix in the SAM framework is very important because it captures the overall impact of changes in one sector of the economy.

Meanwhile, for the SAM-based price model, Roland-Holst and Sancho (1995) and Llop (2021)

			EXPENDITURE			Exogenous Account	TOTAL
			Endogenous Account				
			Production Factors	Institution	Production Activities		
INCOME	Endogenous Account	Production Factors	0	0	T ₁₃	T ₁₄	y ₁
		Institution	T ₂₁	T ₂₂	0	T ₂₄	y ₂
		Production Activities	0	T ₃₂	T ₃₃	T ₃₄	y ₃
	Exogenous Account		T ₄₁	T ₄₂	T ₄₃	T ₄₄	y ₄
	TOTAL		y' ₁	y' ₂	y' ₃	y' ₄	

Figure 2. SAM framework. Source: Hartono and Resosudarmo (1998).

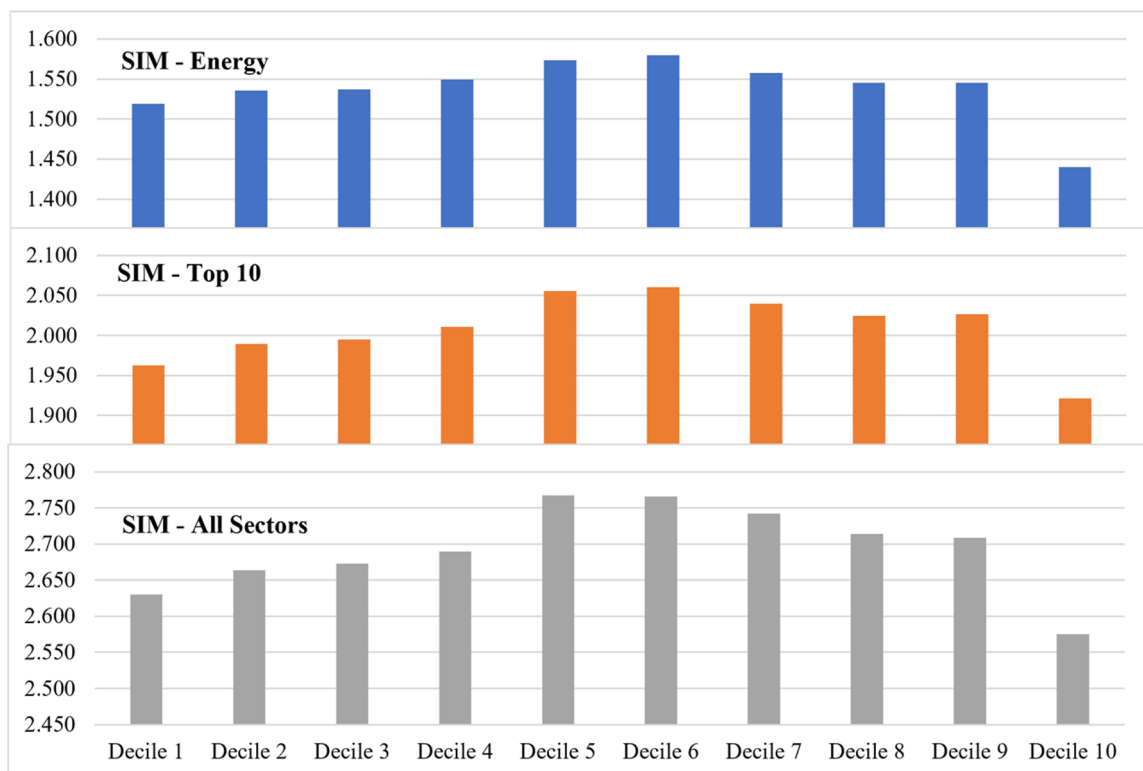


Figure 3. Cost-of-living increase (%) in rural households due to carbon-tax implementation.

denoted a price model, which can be seen in Eqs. (3) to (6). In Eq. (3), $P = p'$, which is the vector of the price index for the endogenous account, A' is the transpose of the coefficient matrix, and v is the vector of the average expenditure coefficient. Modifying Eq. (3), Eq. (4) then shows the magnitude of price increases (ΔP) due to the change in the exogenous account (Δv).

$$P = A'P + v \leftrightarrow P = (I - A')^{-1} \cdot v \leftrightarrow P = M' \cdot v \quad (3)$$

$$\Delta P = M' \cdot \Delta v \quad (4)$$

In the SAM framework with multiple price indexes, Eq. (3) can be restated as Eq. (5), where p_1 represents the price indexes of the factors of production (such as wage rates), p_2 represents the price indexes for household and company expenditures, and p_3 represents the price indexes of sectoral prices. In Eq. (5), the prices in each index depend on the value of the exogenous coefficient v . By modifying Matrix 5, Eq. (6) shows how changes in the exogenous coefficient, in this case, the carbon tax, potentially lead to price increases in the factors of production (Δp_1), household and company expenditures (Δp_2), as well as sectoral prices (Δp_3). Since the carbon tax is applied to sectoral prices $\Delta v_3 \neq 0$, denoting the adoption of the carbon tax (exogenous account) in the sectoral price. Eq. (6) will serve as the reference for the calculations in the analysis.

$$\begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix} = \begin{bmatrix} I & -A_{21} & 0 \\ 0 & I - A_{22} & -A_{32} \\ -A_{13} & 0 & I - A_{33} \end{bmatrix}^{-1} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \quad (5)$$

$$\begin{bmatrix} \Delta p_1 \\ \Delta p_2 \\ \Delta p_3 \end{bmatrix} = \begin{bmatrix} I & -A_{21} & 0 \\ 0 & I - A_{22} & -A_{32} \\ -A_{13} & 0 & I - A_{33} \end{bmatrix}^{-1} \begin{bmatrix} \Delta v_1 \\ \Delta v_2 \\ \Delta v_3 \end{bmatrix} \quad (6)$$

The SAM-based price model used in this study has three main assumptions: (1) The model is a cost-push price model, where prices fluctuate while quantities remain constant, implying that demand is assumed to be perfectly inelastic to price changes (Saari, Dietzenbacher, and Los 2016); (2) price changes in a sector can be transmitted continuously to downstream industries, so the estimated impact might be too high; and (3) the technical coefficient is fixed, so that producers' efforts to reduce costs through technological changes are not taken into account. This model also assumes that SAM always represents an equilibrium condition where the initial price is equal to 1 ($p_{i0} = 1$).

Simulation

We simulated the implementation of a carbon tax at US\$12.00 per ton of emissions, which is equivalent to IDR 183,336.00 (calculated using the Jakarta Interbank Spot Dollar Rate (JISDOR) exchange rate as of May 31, 2021). The tax rate aligns with that used by Nurdianto and Resosudarmo (2016), who examined the short-term effects of a carbon tax in Indonesia set at US\$10 per ton of CO₂. We compared the implementation of a carbon tax across three scenarios: taxing only electricity (SIM-Elect), only the top-10 emitting sectors (SIM-Top 10), and all sectors (SIM-All sectors).

Our scenario focuses on identifying different coverage options because the Indonesian government wants a gradual application. This approach will provide insights into the impact of each step. In addition, while the environmental impact is straightforward—greater coverage means more emissions are addressed—the economic and equity tradeoffs are less clear. For example, addressing twice the number of emissions does not necessarily lead to a doubling of living costs, as the sectors subject to the tax vary, resulting in different indirect tax burdens. In addition, the impact on sectoral and household-income distribution remains unclear. This coverage has also been a subject of discussion in the planning of carbon-tax applications in Indonesia.

The first scenario focuses on implementing a carbon tax within the electricity sector, given its central role in Indonesia's carbon emissions. As one of the largest CO₂ emitters, responsible for nearly 50% of emissions, changes in this sector can directly reduce overall emissions. This policy also aligns with national targets to increase renewable energy in the energy mix, particularly within the electricity sector, and to reduce dependence on fossil fuels. The approach aims to balance emission reduction with economic sustainability, allowing other industries time to adapt and avoiding excessive burdens on sectors already affected by the multiplier effect of electricity usage. The second scenario targets ten key sectors, including other government-priority sectors beyond electricity. These include energy-intensive industries and transportation, which altogether account for over 95% of total national emissions. This approach addresses major emitters while minimizing the impact on sectors like agriculture. Additionally, a broader scenario applies the carbon tax across all economic sectors, covering goods and services that emit carbon through energy use. This scenario demonstrates the effects of full-scale implementation. Previous studies have often focused on energy-intensive sectors and overlooked smaller

emitters, like agriculture. However, with detailed emissions data, we are able to assess the full scope of the impact of the carbon tax.

We simulated two steps, an approach that we adopted from Renner's (2018) study. First, we calculated the price increase in each sector due to the implementation of carbon tax by using the extended IO model. Second, the price increase resulting from the initial step was used as a shock to assess the final and overall impacts of carbon-tax implementation on price increases and to examine the distributional impacts on different household groups, particularly concerning their living costs.

In the first step, we calculated the direct sectoral carbon emission (C_{dir}) from energy consumption by multiplying the carbonization factor per unit of fuel used (e_f) by the total fuel used by a sector (f) (see Eq. (12)). We then calculated the direct sectoral CO₂ emission intensity (CI_{dir}) by dividing C_{dir} by the number of sectoral outputs in the IO 2015 Table (see Eq. (13)). Next, we calculated the total sectoral CO₂ emission intensity (CI_{tot}) (see Eq. (14)) by multiplying CI_{dir} matrix with the Leontief inverse matrix as previously done by Renner (2018). Finally, the ad

valorem carbon-tax rate (t) that must be paid by each sector can be calculated by multiplying the carbon-tax tariff (μ) with the total sectoral CO₂ intensity (CI_{tot}) (see Eq. (15)).

$$C_{dir} = e'f \quad (12)$$

$$CI_{dir} = \frac{C_{dir}}{X_i} \quad (13)$$

$$CI_{tot} = CI_{dir} (I - A)^{-1} \quad (14)$$

$$t = \mu * CI_{tot} \quad (15)$$

$$(1 + t_i) p_{i0} = p_{i1} \leftrightarrow (1 + t_i) = \frac{p_{i1}}{p_{i0}} \leftrightarrow t_i = \frac{p_{i1}}{p_{i0}} - 1 \quad (16)$$

The amount of ad valorem carbon tax (see Eq. (16)) that a sector must pay could be considered as the change in the relative price of a sector to its base price (p_{i0}). Thus, the base price of each sector is assumed to be 1 ($p_{i0} = 1$).

Table 2. Sectoral direct CO₂ emissions (C_{dir}), direct CO₂ intensity (CI_{dir}), and total CO₂ intensity (CI_{tot}).

No. (1)	Sector (2)	Direct CO ₂ Emissions (ton) (3)	Direct CO ₂ Intensity (tons/IDR million) (4)	Total CO ₂ Intensity (ton/IDR million) (5)
1	Food Crops	1,204.43	0.000	0.004
2	Other Crops	117,501.63	0.000	0.007
3	Livestock	25,068.13	0.000	0.011
4	Forestry	80,696.68	0.001	0.005
5	Fishery	217,037.92	0.001	0.005
6	Coal and Metal Mineral	433,589.25	0.001	0.009
7	Crude Oil and Natural Gas	3,628,701.54	0.007	0.014
8	Other Mining	403,511.45	0.001	0.009
9	Food and Beverage	3,005,543.68	0.001	0.016
10	Textile and Leather	2,388,555.46	0.005	0.044
11	Timber and Wooden Products	1,636,350.88	0.006	0.033
12	Paper	1,638,955.00	0.003	0.030
13	Basic non-Ferrous	6,955,822.86	0.022	0.048
14	Metal Product	24,540,457.75	0.033	0.059
15	Boiler and Prime Mover	554,262.90	0.028	0.071
16	Electricity Machinery and Equipment	1,297,852.34	0.001	0.027
17	Chemical	55,433,220.97	0.034	0.050
18	Rubber and Plastic	8,622,008.62	0.041	0.062
19	Electricity	224,999,355.17	0.558	1.291
20	Gas and Clean Water	790,567.38	0.014	0.033
21	Power Building Construction	16,215,701.95	0.023	0.043
22	Other Construction	61,090,334.25	0.023	0.044
23	Trading	7,146,366.55	0.003	0.015
24	Restaurant	48,041.36	0.000	0.022
25	Hotels	18,860.11	0.000	0.025
26	Rail Transportation	231,575.69	0.014	0.085
27	Other Land Transportation	18,353,982.45	0.032	0.044
28	Air and Water Transportation	22,098,414.31	0.017	0.033
29	Transportation Supporting Services	532,400.70	0.003	0.019
30	Bank and Insurance	284,803.60	0.000	0.010
31	Real Estate	51,143.06	0.000	0.007
32	Company Service	332,738.14	0.001	0.014
33	General Government and Defense	4,309,081.67	0.003	0.017
34	Social Community Services	56,094.14	0.003	0.008
35	Other Services	352,149.77	0.002	0.006

Sources: Author's calculations (2021) using data from Statistics of Indonesia Energy Balance (2015).

Table 3. Sectoral ad valorem tax rate (%) and price increases (%).

No. (1)	Sector (2)	Ad valorem tax rate (%) (3)	Price increases (%) (4)		
			SIM-Elec	SIM-Top 10	SIM-All sectors
1	Food Crops	0.066	0.986	1.351	1.865
2	Other Crops	0.137	1.092	1.573	2.205
3	Livestock	0.195	1.223	1.669	2.411
4	Forestry	0.098	1.023	1.407	1.967
5	Fishery	0.099	1.050	1.410	1.962
6	Coal and Metal Mineral	0.166	0.991	1.422	2.054
7	Crude Oil and Natural Gas	0.255	0.938	1.320	1.968
8	Other Mining	0.164	1.046	1.495	2.142
9	Food and Beverage	0.302	1.267	1.716	2.539
10	Textile and Leather	0.808	2.076	2.559	3.959
11	Timber and Wooden Products	0.598	1.534	2.094	3.265
12	Paper	0.541	1.473	2.098	3.153
13	Basic non-Ferrous	0.881	1.365	2.917	3.409
14	Metal Product	1.075	1.374	3.121	3.590
15	Boiler and Prime Mover	1.297	2.034	2.493	4.194
16	Electricity Machinery and Equipment	0.501	1.509	2.072	3.058
17	Chemical	0.913	1.078	2.457	2.888
18	Rubber and Plastic	1.144	1.423	3.079	3.576
19	Electricity	23.660	55.325	56.143	56.931
20	Gas and Clean Water	0.604	1.444	1.872	2.937
21	Power Building Construction	0.789	1.104	2.554	3.054
22	Other Construction	0.805	1.133	2.614	3.120
23	Trading	0.279	1.225	1.960	2.471
24	Restaurant	0.396	1.594	2.053	3.060
25	Hotels	0.464	1.736	2.212	3.188
26	Rail Transportation	1.562	3.565	4.285	6.527
27	Other Land Transportation	0.800	1.096	2.346	2.931
28	Air and Water Transportation	0.612	1.179	2.348	2.865
29	Transportation Supporting Services	0.345	1.246	1.857	2.710
30	Bank and Insurance	0.180	1.217	1.652	2.344
31	Real Estate	0.132	0.906	1.332	1.850
32	Company Service	0.255	1.098	1.616	2.375
33	General Government and Defense	0.316	1.502	2.074	3.013
34	Social Community Services	0.154	0.883	1.216	1.792
35	Other Services	0.119	1.420	1.854	2.501

Sources: Author's calculations (2021) based on data from Statistics of Indonesia Energy Balance (2015).

Results

Since all production sectors in the economy use different amounts of energy input, the amount of direct CO₂ emissions (C_{dir}) of each sector is different from other sectors (see the C_{dir} in Table 2). After calculating C_{dir} for each sector, we calculated the direct CO₂ emissions intensity (CI_{dir}) according to Eq. (13), so that CI_{dir} could be obtained (see Table 1). After getting the value of each sector CI_{dir} , we calculated the total CO₂ emissions intensity (CI_{tot}) by multiplying CI_{dir} with the Leontief inverse matrix according to Eq. (14) so that the value of CI_{tot} could be known (see Table 2).

After obtaining the value of total CO₂ emissions intensity (CI_{tot}) for each sector, we calculated the amount of the ad valorem tax rate (in % of sectoral output) that must be borne and paid by each sector (see Table 3). Later, this tax rate is considered as a relative price change to the initial price, where the initial price is assumed to be equal to 1 ($p_{i0} = 1$). Then, we calculated the price increase for each sector in SIM-Elec, SIM-Top 10, and SIM-All sectors using the price multiplier matrix. The price increases can be seen in Table 3.

In 2015, the electricity sector emerged as the leading contributor to CO₂ emissions, accounting for over 224 million tons of emissions (see Table 2). Outside the manufacturing sector, other significant emitters included the construction sector, which contributed 61.1 million tons of emissions, and various transportation sectors, including air and water transportation (22.1 million tons) and land transportation, excluding rail transportation (18.4 million tons). Within the manufacturing sector, the top emitters were the chemicals sector, responsible for 55.4 million tons of emissions, and the metal-products sector, emitting 24 million tons. The remaining sectors within manufacturing each produced less than 10 million tons of emissions.

Aside from being the top emitter, the electricity sector exhibited the highest emission intensity, both in terms of direct emissions and overall emissions. Given that a substantial portion of Indonesia's power plant relies on coal, it is expected that the electricity sector would also possess a high level of direct and total emission intensity, reaching 0.558 and 1.291, respectively, while on average, the direct emission intensity is 0.025 and the total emission intensity is

0.063 for all sectors. Compared to the average emissions, certain top-emitter sectors, such as construction (0.032) and chemicals (0.034), also exhibit relatively high direct emissions. In addition, it is worth noting that several sectors may not have a high direct emission intensity but display a high total emission intensity. Examples include rail transportation with a total emission intensity of 0.085 and the boiler and prime mover sector with a total emission intensity of 0.071. Yet these values are still greatly surpassed by the electricity sector.

Table 2 also shows the amount of emission subject to tax in each scenario. SIM-elect imposes taxes on 224 million tons of emissions, SIM-Top 10 imposes taxes on 445 million tons of emissions (nearly double the amount in SIM-Elec), and SIM-All sectors impose taxes on 467 million tons of emissions. The top-10 sectors responsible for the highest emissions include electricity generation, non-ferrous metal production, chemical manufacturing, rubber and plastic industries, power building construction, other construction, trading, other land transportation, and air and water transportation. These are the sectors that will be subject to taxation under the SIM's top-10 sectors.

As presented in Table 3, the highest ad valorem tax rate is observed in the electricity sector at 23.60%. This is followed by the rail-transportation sector at 1.56%, boiler prime mover at 1.30%, and rubber and plastic at 1.14%. The electricity sector significantly contributes to overall tax revenue,

surpassing all other sectors by a considerable margin. The imposition of this tax leads to a substantial increase in the price of electricity, ranging from 55% to 57% across various simulation scenarios, in stark contrast to other sectors that experience relatively minor price increases of around 0% to 4%. It is important to note that all sectors see price hikes due to increased production costs stemming from rising energy prices after the implementation of the carbon tax. Additionally, labor and non-labor input prices also experience increases (See Table 4).

Different scenarios have varying impacts on prices across sectors. Under the “imposing tax in only the electricity sector” (SIM-Elec) scenario, sectors such as rail transportation (3.56%), boiler and prime mover (2.03%), and textile and leather industries (2.08%) witness the highest price increases after the electricity sector. In contrast, other sectors see price increases of less than 2%. Under the top-10 scenario (SIM-top 10), a broader range of sectors experience price hikes exceeding 2%, including all manufacturing industries, transportation sectors, and restaurants and hotels, with increases ranging from approximately 2% to 3%. In the scenario where tax is imposed across all sectors (SIM-All), the price increases are consistently around or above 2%, with all manufacturing sectors experiencing increases of more than 3%. Notably, the imposition of tax across all sectors results in food crops, other crops, and the food and beverage sectors seeing price increases exceeding 2%, while in other simulations, these

Table 4. Factor-price increases (%) due to carbon-tax implementation.

					Increase in factor prices (%) due to SIM-Energy	Increase in factor prices (%) due to SIM-Top 10	Increase in factor prices (%) due to SIM-All Sectors
Classification of production factors							
Labor	Agricultural	Formal	Rural	1.539	2.009	2.694	
			Urban	1.756	2.268	2.931	
		Informal	Rural	1.542	2.011	2.697	
			Urban	1.779	2.284	2.954	
	Manual	Formal	Rural	1.531	2.006	2.689	
			Urban	1.723	2.245	2.892	
		Informal	Rural	1.536	2.009	2.694	
			Urban	1.757	2.271	2.931	
	Clerical	Formal	Rural	1.514	1.992	2.669	
			Urban	1.681	2.216	2.850	
		Informal	Rural	1.532	2.006	2.690	
			Urban	1.739	2.259	2.912	
	Professional	Formal	Rural	1.499	1.978	2.651	
			Urban	1.642	2.179	2.800	
		Informal	Rural	1.514	1.990	2.668	
			Urban	1.667	2.209	2.840	
Non-Labor				0.823	1.084	1.443	

Sources: Author's calculations (2021) using data from Statistics of Indonesia Energy Balance (2015).

Table 5. The impact of carbon-tax implementation on the cost-of-living index due to Simulation 1 and Simulation 2.

Household's Geographical Location		Household Groups	Cost of Living Index Increases (%) due to SIM-Energy	Cost of Living Index Increases (%) due to SIM-Top 10	Cost of Living Index Increases (%) due to SIM-All Sectors
Rural	Decile 1	Decile 1	1.519	1.962	2.630
		Decile 2	1.536	1.989	2.664
		Decile 3	1.537	1.995	2.673
		Decile 4	1.549	2.011	2.690
		Decile 5	1.573	2.056	2.767
		Decile 6	1.580	2.060	2.766
		Decile 7	1.558	2.039	2.742
		Decile 8	1.545	2.025	2.714
		Decile 9	1.545	2.026	2.708
		Decile 10	1.440	1.921	2.575
Urban	Decile 1	Decile 1	1.790	2.275	2.964
		Decile 2	1.815	2.308	3.003
		Decile 3	1.805	2.302	2.984
		Decile 4	1.795	2.296	2.962
		Decile 5	1.799	2.307	2.963
		Decile 6	1.767	2.286	2.948
		Decile 7	1.731	2.258	2.898
		Decile 8	1.691	2.225	2.856
		Decile 9	1.655	2.213	2.837
		Decile 10	1.570	2.135	2.740

Sources: Author's calculation (2021) using data from Statistics of Indonesia Energy Balance (2015).

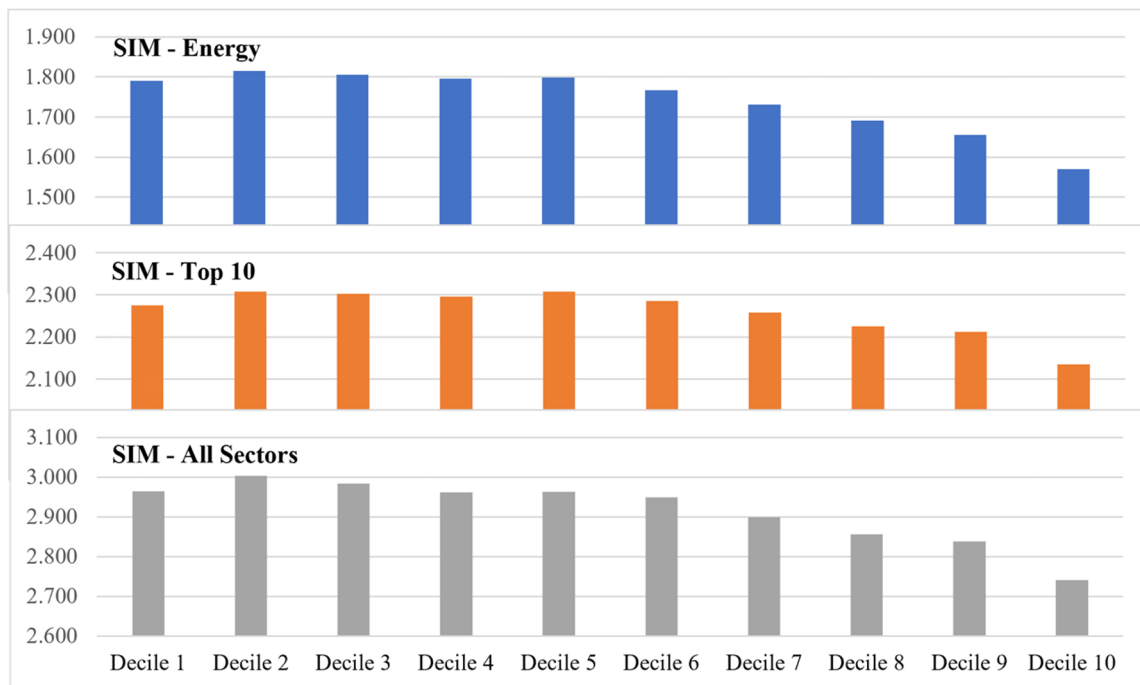


Figure 4. Cost-of-living increase (%) in urban households due to carbon-tax implementation.

sectors only experience approximately 1% price increases. This is a significant consideration, as price increases in the food sector, which produces essential household necessities, can potentially impact vulnerable households by pushing them closer to poverty.

From Tables 4 and 5, it is evident that imposing a tax will raise both the price of labor (or, in other words, labor income) and the cost of living. Across all labor types and all household groups, the simulation encompassing all sectors appears to have a more pronounced impact in terms of rising living expenses and labor costs compared to the simulation focused solely on the top-10 emitting sectors, particularly when considering that the emissions subject to taxation in SIM-All sectors is only slightly higher as opposed to the top-10 sectors. This difference is even more striking when contrasted with the impact of the SIM focused solely on the electricity sector (SIM-Elec). The magnitude of this impact gap between the SIM covering all sectors and the SIM targeting the top 10 sectors ranges from approximately 0.5% to 0.7%, whereas the gap between SIM-Elect and SIM-Top 10 sectors falls within the range of 0.3% to 0.5%.

Examining the equity between urban and rural areas, labor-price increases are relatively higher in urban areas than in rural areas. On average, urban labor incomes increased by 0.21% more than those of rural households. However, urban households face more significant challenges due to a rise in the cost of living compared to rural households at the same decile level (see Table 5). On average, urban households experience a 0.23% higher increase in living costs than rural households in the same decile,

indicating that the rise in living costs outpaces the increase in income. This pattern persisted across all simulations with similar gap values.

Regarding the effect across different income levels, the findings revealed distinct patterns between rural and urban households. In rural settings (see Figure 3), the increase in the cost-of-living index from Decile 1 to Decile 10 follows an inverted-U pattern, indicating regressive effects in the approximately 60th to 100th percentile range. This means that in rural areas, middle-income people suffer the most from the carbon tax. Meanwhile, urban households also displayed an inverted U pattern, although it was not as pronounced as in rural areas (see Figure 4). Initially, there is an increase in the cost-of-living index from Decile 1 to Decile 2, followed by a regressive pattern. This suggests that the bottom 20% are the most affected by the carbon tax in urban areas. Interestingly, the top 10% in both urban and rural areas, as well as in all scenarios, are the least burdened by the carbon tax. Such a regressive pattern due to carbon tax was also found in Mexico by Chapa and Ortega (2017), but it differs from a finding found by Yusuf and Resosudarmo (2015). Meanwhile, comparing the scenarios within rural households, the inverted U pattern is more apparent in the scenario involving all sectors. In contrast, within urban areas, all scenarios exhibit similar patterns.

Discussion

The analysis shows that the electricity sector is the largest contributor to CO₂ emissions and bears the highest ad valorem tax rate at 23.6%. Other than

electricity, significant emitters include the construction sector and various transportation sectors, while within manufacturing, the chemical and metal-product sectors are top emitters. High emission intensities are also noted in sectors like rail transportation and the boiler and prime mover sectors. Using the SAM-based price model, we found that overall, the implementation of a carbon tax will lead to a significant rise in the price of electricity, ranging from 55% to 57% across various simulation scenarios. In comparison, other sectors will experience relatively minor price increases of around 0% to 4%. The price hikes are mainly due to rising production costs from increased energy prices. Additionally, labor and non-labor input prices will also experience increases.

Regarding the distributional impacts of carbon-tax implementation among households, urban households face greater challenges due to a more significant increase in living costs than rural households at the same income decile. This disparity arises from distinct consumption patterns: rural households allocate around 31.7% of their income to food, which is less affected by the carbon tax. In contrast, urban households spend relatively less on food but more on energy-intensive sectors like transportation and electricity. A study on the carbon tax in the Philippines also found that urban households bear a higher burden than rural households (Cabalu et al. 2015). This suggests that a carbon tax may help reduce the gap between rural and urban households.

Comparing the scenarios, the results reveal varying consequences of different carbon-tax implementations. Environmentally, the scenario covering all sectors (SIM-All) provides the highest benefit, covering 467 million tons of emissions, which is 208% more than when the tax is applied only to the electricity sector (SIM-Elec). However, SIM-Top 10, which targets the top-10 emitting sectors, covers 445 million tons of emissions, or 190% more than SIM-Elec. Although imposing the tax on all sectors provides the greatest environmental benefit, the difference in emission coverage between taxing all sectors and only the top-10 emitting sectors is relatively small.

Economically, imposing the tax only on the electricity sector primarily affects electricity prices, resulting in modest price shifts in other sectors and relatively contained economic disruption. Expanding the tax to the top-10 emitting sectors extends these effects to a broader range, including manufacturing and transportation, with price increases around 2–3%. However, the SIM-All scenario, despite only a marginal increase in emission coverage compared to SIM-Top 10, results in significant price increases across all sectors, including food-related sectors.

This scenario imposes substantial cost burdens on living expenses and labor, leading to the most widespread economic disruption by extensively affecting both production costs and consumer prices.

In terms of equity, all scenarios show similar patterns. The carbon tax exhibits an inverted U-shaped regressivity for rural households and overall regressivity for urban households. This means that middle-income households in rural areas and the bottom 20% in urban areas suffer the most compared to other categories. However, imposing the carbon tax across all sectors (SIM-All) exacerbates inequity across household incomes compared to other scenarios, particularly within rural households.

The discussion on the regressivity effect from prior studies varies. Chen and Nie (2016), Metcalf (2021), and most research indicate that carbon taxes in developed countries have regressive effects while developing countries show faint progressive effects. Our study shows that even within a country, the regressive pattern may not be uniform. Our analysis reveals that total regressivity is only found in urban areas. In rural areas, we found an inverted U-shaped pattern, where the effect is progressive at lower income levels and becomes regressive at higher income levels.

This inverted U-shaped pattern in rural areas can be explained by the relationship between income and energy expenditure. At lower income levels, households spend a larger proportion of their income on essential goods and services and less on energy-based products. As their income increases, they diversify their spending and can afford more products, including energy-intensive items such as motorcycles and electronic appliances. However, the highest income groups can afford more energy-efficient technologies, reducing the relative burden of the carbon tax. Consequently, despite overall price hikes due to increased production costs, the top 10% in all scenarios remain the least burdened by the carbon tax.

Additionally, low-income rural households in Indonesia often rely more heavily on traditional biomass for energy, which is typically exempted from carbon pricing. This exemption may also explain the progressive impact at lower income levels, as argued by Dorband et al. (2019) and Steckel et al. (2021). As incomes rise and households transition to more modern energy sources, the tax's regressive effects become more pronounced. However, the total regressive impact is more evident in urban households, where energy is a necessary good for all income levels. This insight is also supported by Dorband et al.'s (2019) findings, which suggest that there is a turning point in household income where the

progressive effect pattern becomes regressive. Since household incomes in Indonesia are clustered around this turning point, the distributional effects are less clear. To conclude, this finding aligns with the broader literature on the distributional impacts of carbon pricing, which suggests that the effects can vary significantly based on local economic conditions and energy-consumption patterns (Dorband et al. 2019; Steckel et al. 2021). It underscores the importance of designing carbon-tax policies that consider these variations to mitigate adverse impacts on vulnerable populations.

Based on the analysis, we consider several policy implications. First, the electricity sector plays a significant role in Indonesia, similar to other Southeast Asian countries (Ruamsuke, Dhakal, and Marpaung 2015). Therefore, implementing clean technology in this sector is crucial for emissions reductions and can have positive economic impacts under stringent emissions caps.

Second, considering economic, environmental, and equity aspects, it is advisable to impose a carbon tax on the electricity sector or the top-10 emitting sectors. However, extending the tax to all sectors is not recommended. Our analysis suggests that the difference in emission coverage between SIM-All and SIM-Top 10 is relatively small, but imposing the tax on all sectors results in significant price increases, including in food-related sectors. This could push vulnerable populations closer to poverty.

Finally, it is essential in Indonesia to implement policies that alleviate the burden on middle-income households in rural areas and lower-income households in urban areas, as these groups are the most affected by the carbon tax. It is also important to note that while carbon pricing is currently not regressive, there is a potential for regressivity in rural areas if low-income households gain more access to energy-related sectors. Therefore, additional policies will be necessary to prevent adverse impacts on the poor.

The Indonesian government could consider several complementary policies to mitigate the adverse effects of a carbon tax. One potential intervention is to introduce a tax-shifting package, where carbon-tax revenues are used to reduce income taxes by a certain percentage. However, since not all households bear the same burden from the carbon tax, this measure may only partially offset the negative impact on welfare, as evidenced by studies in Finland (Khastar, Aslani, and Nejati 2020). Alternatively, we recommend schemes such as lump-sum dividends, government provision of public services, or targeted financial transfers to households—especially the lowest-income households in urban areas and

middle-income households in rural areas. These measures can reduce the adverse effects of the carbon tax and lessen inequality across different income levels. Another policy option is a recycling scheme for carbon-pricing revenues to support business activities, such as investment and tax reductions, which can help reduce opposition from business stakeholders (Dyarto and Setyawan 2021). Implementing these complementary policies can enhance the political feasibility and social acceptability of the carbon tax, ensuring that the transition to a low-carbon economy is equitable and sustainable.

Conclusion

Using the SAM price model, this study attempted to identify the effect of carbon-tax adoption on price increases and distributional impacts. The analysis indicates that a carbon tax will substantially increase electricity prices and moderately impact other sectors, especially rail transportation and electricity machinery and equipment. Distributionally, urban households bear a heavier burden due to higher living costs, while rural households exhibit an inverted U-shaped regressivity pattern. Different carbon-tax scenarios show that imposing the tax on all sectors provides the greatest environmental benefit but results in moderate price increases for transportation and food-related sectors compared to other scenarios, potentially pushing vulnerable populations into poverty. Therefore, we advise targeting the tax on the electricity and top-10 emitting sectors but do not recommend it for all sectors. Complementary policies such as lump-sum dividends, public services, and targeted financial transfers are necessary to mitigate the adverse impacts on middle-income rural and lower-income urban households. These measures will enhance the carbon tax's political feasibility and social acceptability, ensuring a more equitable and sustainable transition to a low-carbon economy.

While this study provides insights into the impact of carbon-tax implementation on price increases and distributional effects, it also opens several avenues for future research. First, our analysis focused on a national context, suggesting a potential area for future studies to explore regional impacts and differences in more detail. Second, the assumptions used in the SAM model, such as the Leontief production function, did not account for the substitution effect, leading to a direct impact on consumers. Future research could investigate alternative production functions that might better capture these dynamics. Third, the SAM model's current application is more valid for short-term impact simulation. The model's accuracy may decrease as tax rates increase and the

time horizon extends. Finally, we conducted this study using data from 2015, which did not account for changes in the economic structure due to the COVID-19 pandemic. Future research should consider more recent data to understand better the ongoing and future economic impacts of such global disruptions.

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No potential conflict of interest was reported by the authors.

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